Oluwatosin Ayeni

Staff/Principal Machine Learning Engineer (Serving/Inference) — Inworld AI

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PROFILE

Production-focused ML engineer with deep hands-on experience in LLM inference optimisation, GPU serving infrastructure, and multimodal AI systems. Benchmarked vLLM v0.30 and SGLang v0.5.20 on dual NVIDIA H100 NVL 94GB — 5,965 tok/s throughput and 1,443× TTFT speedup via RadixAttention at 131K-token context. Delivered 28× cost reduction via continuous batching and KV-cache optimisation. Built multimodal inference lab covering all 5 output modalities. UK Remote.

CORE COMPETENCIES

Inference Serving	vLLM, SGLang, TensorRT-LLM, LiteLLM — multi-modal serving pipelines (text, audio, image, structured output)
GPU / Systems	CUDA kernel optimisation, NVIDIA Nsight, KV-cache tuning, speculative decoding, continuous batching, quantisation (AWQ)
MLOps / K8s	Kubernetes, KEDA autoscaling, Prometheus/Grafana/Loki, HPA, Velero backup, RunPod, Vast.ai
LLM / Frameworks	PyTorch, PEFT/LoRA fine-tuning (Mistral 7B, Phi-4, Qwen3, Gemma 2), RAG, LangChain, Claude API
Engineering	Python, FastAPI, Docker, Redis, PostgreSQL, n8n, GitHub Actions, Next.js, TypeScript

PROFESSIONAL EXPERIENCE

Jan 2025–present — **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present — **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present — **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present — **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 — **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

KEY PROJECTS

llm-inference-benchmark (H100) vLLM v0.30 vs SGLang v0.5.20, 2× H100 NVL 94GB, Qwen3-8B BF16 TP=2. RadixAttention: 1,443× TTFT speedup. Peak: 5,9 tok/s. 100% SLO at concurrency 128. Live results

lab-vllm-omni vLLM-Omni phased inference lab: all 5 output modalities — text, audio, image, video, structured output. GitHub

llm-inference-platform Production serving on RunPod and Vast.ai: vLLM + SGLang + LiteLLM gateway, KEDA autoscaling, Prometheus/Grafana/Loki. GitHub

inference-optimizer Production Inference Gateway: 3.87× speed, 61.4% cost reduction, 64.3% cheaper routing, 14× throughput. GitHub

open-model-engine GPU-first inference benchmarking: CUDA profiling, quantization, speculative decoding. GitHub

EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

OPEN SOURCE

GitHub 43 public repos — github.com/AyeniOluwatosinOlawale